AXOLABS METHOD VALIDATION REPORT FOR THE ANALYSIS OF 2439-EXNA IN DOG PLASMA	
Axolabs Project Number	AN402.0-24
Project Title	Validation of an AEX-HPLC-based method for the analysis of 2439-exNA in dog plasma
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Start Date of Bioanalytical Test Phase	06-Sep-2024
End Date of Bioanalytical Test Phase	18-Sep-2024

SIGNATURE PAGE

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Alissa Coffey **SPONSOR'S REPRESENTATIVE**	Signed by: [redacted] Signer Name: Alissa Coffey Signing Reason: I approve this document Signing Time: 04-Dez-2024 | 14:47 CET 3E846AC45A5F4A42B5DCE31B3306D2F1

COMPLIANCE STATEMENT

This *Validation of an AEX-HPLC-based method for the analysis of 2439-exNA in dog plasma* was conducted in accordance with the procedures described herein.

The method validation was conducted in compliance with *ICH guideline M10 on bioanalytical method validation and study sample analysis* (EMA/CHMP/ICH/172948/2019) and following the rules in the *OECD Principles of Good Laboratory Practice* (Revision from 1997).

There were no deviations from the Method Validation Plan and respective SOPs. All work was done according to the Method Validation Plan and respective SOPs.

The Method Validation Report represents an accurate and complete record of the results obtained during the method validation.

No deviations from the above regulations affected the overall integrity of this method validation.

Study Director:	Nicole Rottmann	Signiert von: [redacted] Name des Unterzeichners: Nicole Rottmann Signiergrund: Ich habe dieses Dokument geprüft Signierzeit: 04-Dez-2024 | 08:57 MEZ 87289BA18FC048139E8770AAAEB8DA48
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QUALITY ASSURANCE STATEMENT

This Method Validation Report titled *Validation of an AEX-HPLC-based method for the analysis of 2439-exNA in dog plasma* has been audited.

The reported results accurately reflect the raw data generated during this method validation.

As the bioanalytical testing is of short duration, routine and repetitive, Axolabs GmbH applies a process-based inspection program. The process-based inspection relevant to the activities performed during this method validation is listed below.

Table 1: Quality Assurance Inspections

Type(s) of Inspection(s)	Inspected Activities	Date(s) of the Inspection(s)	Date(s) of Reporting and Recipient(s)
Process-based (Report No.: II-24-003)	• Incoming Goods Check, storage, and labeling of study samples • Incoming Goods Check, storage, and labeling of reference standard, probe, blank matrix • Labeling and preparation of reagents/ solutions • Labelling and preparation of reference standard stock solution • Preparation of calibration standards and QC samples • Sample preparation • Lysate preparation and hybridization • HPLC setup and analysis • HPLC data processing and data evaluation • Disposal of study samples	03, 05, 11 & 18-Sep-2024	TSM: 11-Oct-2024 SD: 15-Oct-2024
Audit of Report and Data	Draft Report Version from 21-Nov-2024	27-Nov-2024	SD: 27-Nov-2024 TSM: 03-Dec-2024

TSM: Test Site Management. SD: Study Director.

In addition to the listed process-based inspections, facility-based inspections take place yearly.

Quality Assurance Axolabs:	Anne Linhardt	Signiert von: [redacted] Name des Unterzeichners: Anne Linhardt Signiergrund: Ich habe dieses Dokument geprüft Signierzeit: 04-Dez-2024 | 10:10 MEZ E915197353544427B9CF2FC3E20B813C
	Name	Date and Signature

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LIST OF ABBREVIATIONS

AB0X	Analysis Batch, AN402.0-24_01_AB0X
ACN	Acetonitrile
AEX-HPLC	Anion-Exchange High-Performance Liquid Chromatography
BQL	Below Quantification Limit (Below the LLOQ)
CHMP	Committee for Human Medicinal Products
CoA	Certificate of Analysis
Conc.	Concentration
CV	Coefficient of Variation
DPBS	Dulbecco´s Phosphate Buffered Saline
DNA	Deoxyribonucleic Acid
EMA	European Medicines Agency
EU	European Union
FTS	Freeze-Thaw Stability
GCP	Good Clinical Practice
GLP	Good Laboratory Practice
ICH	International Council for Harmonisation of Technical Requirements for Pharmaceuticals for Human Use
K_2-EDTA	Dipotassium Ethylenediaminetetraacetic Acid
LTS	Long-Term Stability
LLOQ	Lower Limit of Quantification
MDR	Method Development Report
n.a.	Not Applicable
OECD	Organization for Economic Co-operation and Development
PNA	Peptide Nucleic Acid
PrP	Prion Protein
PSS	Processed Sample Stability
QA	Quality Assurance
QC	Quality Control
SD	Standard Deviation
SDS	Sodium Dodecyl Sulfate
siRNA	small interfering Ribonucleic Acid
SOP	Standard Operating Procedure
SST	System Suitability Test(ing)
STM	Standard Test Method
STS	Short-Term Stability
ULOQ	Upper Limit of Quantification
UV	Ultraviolet

1 SUMMARY

This report summarizes the results of the bioanalytical method validation for the analysis of 2439-exNA in dog plasma (Axolabs Project: AN402.0-24).

The assay is based on the specific hybridization of the antisense strands of 2439-exNA, a dimer composed of two identical siRNAs, with a complementary peptide nucleic acid (PNA) strand in a 1:1 stoichiometry. The PNA is labeled with an Atto425 fluorescence dye and yields a specific signal in the subsequent analysis by Peptide Nucleic Acid Anion-Exchange High-Performance Liquid Chromatography (PNA-AEX-HPLC) coupled to a fluorescence detector.

The method was developed without the need for any analyte extraction from dog plasma. Therefore, recovery of 100 % is maintained over the process (no internal standard is needed).

The bioanalytical method was previously developed at Axolabs GmbH and is described in the Axolabs Method Development Report (MDR) *MDR_2439exNA_DogPlasma* (edition 02, effective date: 20-Aug-2024) that was used as the basis for this method validation (Supplemental File 1). During the method development phase, the suitability of the method for dog plasma was shown.

The following parameters described in the MDR had to be adjusted/adapted in the method validation:

- In the method development a lyophilized reference standard was used, which was weighed and dissolved. In the validation, an already prepared reference standard solution was used, which was provided by the Sponsor.
- Deviating from the MDR, the working solutions, used for the evaluation of LTS, FTS and STS were prepared in glass vials (instead of Eppendorf Protein LoBind tubes) and 0.1% SDS (instead of dog plasma) was used as diluent for the working solutions to avoid non-specific binding of the analyte to the sample vial.

The final conditions of the validated method are described in the Standard Test Method (STM) *STM_Broad_2439-exNA_DogPlasma* (revision 01, effective date: 04-Oct-2024; Supplemental File 2).

The work was done according to the Method Validation Plan signed on 05-Sep-2024 by the Sponsor's Representative.

The results of the method validation are summarized in Table 2.

Table 2: Overall Summary Table

Items	Results	Section/Table	Comments
Analyte	2439-exNA	Section 3	n.a.
Reference Standard	2439-exNA	Section 4	n.a.
Methodology	PNA-AEX-HPLC-Assay with Fluorescence Detection	Section 8.1	n.a.
Biological Matrix (Anticoagulant)	Dog plasma (K_2-EDTA)	Section 5 Table 3	n.a.
Calibration Curve Range	1.0 ng/mL – 1,000.0 ng/mL	Section 9.2 Table 9	n.a.
Selectivity	5 times higher response (peak area) in the spiked QC-LLOQ samples compared to the signal in the blank samples	Section 9.3 Table 11	n.a.
Matrix Effect	No matrix effect observed for 7 (6 non-hemolyzed / non-lipemic and 1 hemolyzed matrix lot) individually tested matrix lots	Section 9.4 Table 12	n.a.
Carry-Over	<20 % of the response (peak area) of the LLOQ calibration standard.	Section 9.5 Table 13	n.a.
Between-Run Accuracy	QC-LLOQ (1.0 ng/mL): 97.2 % QC-Low (3.0 ng/mL): 98.0 % QC-Mid (450.0 ng/mL): 99.7 % QC-High (900.0 ng/mL): 99.0 %	Section 9.7.2 Table 14	n.a.
Between-Run Precision	QC-LLOQ (1.0 ng/mL): 5.9 % QC-Low (3.0 ng/mL): 2.4 % QC-Mid (450.0 ng/mL): 1.5 % QC-High (900.0 ng/mL): 0.7 %	Section 9.8.2 Table 14	n.a.
Dilution Integrity	Dilution QC: 500,000.0 ng/mL (Dilution factor: 1,000) Accuracy: 95.2 % Precision: 0.7 %	Section 9.9 Table 15	n.a.
Long-Term Stability	Ongoing (T=0) at -80 °C ± 20 °C	Section 9.10.1 Table 16	New preparation of stability samples (for LTS, FTS and STS) under AB03.
Freeze-Thaw and Short-Term Stability	Demonstrated for 5 cycles at -80 °C ± 20 °C and for 4 hours on ice	Section 9.10.2 Table 17	
Bench-Top Stability (Robustness)	Was not performed as test was covered by the short-term stability.		
Processed Sample Stability	Demonstrated for 77 hours at 20 °C ± 3 °C	Section 9.10.4 Table 18	n.a.
Short-Term Stability of Stock and Working Solution	Demonstrated for 4 hours at room temperature	Section 9.10.5.1 Table 19	n.a.
Long-Term Stability of Stock Solution	Was not performed as data were provided by the Sponsor.		

2 COMPLIANCE

The method validation was conducted in compliance with the following guidelines, regulations, or guidance documents:

- ICH guideline M10 on bioanalytical method validation and study sample analysis (EMA/CHMP/ICH/172948/2019); European Medicines Agency; 21 January 2023

Additionally, this method validation project was conducted following the rules as depicted in the *OECD Principles of Good Laboratory Practice* (Revision from 1997).

3 ANALYTE

The analyte 2439-exNA is a prion protein (PrP)-lowering divalent siRNA (Figure 1). The sense strands of the two identical siRNAs are linked to each other via a tetraethylene glycol linker.

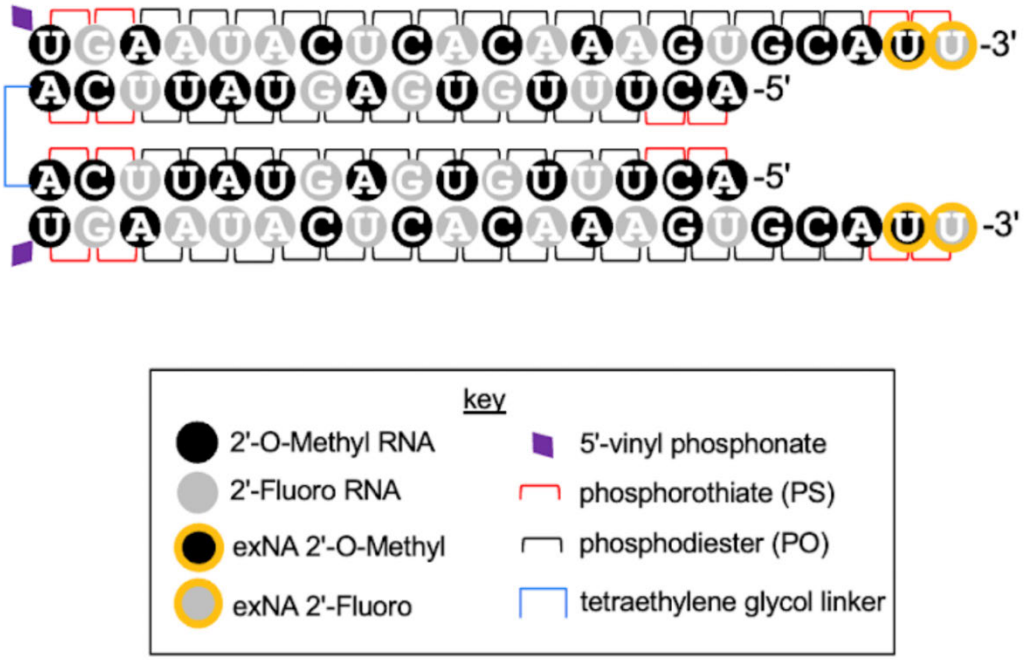


Figure 1: Structure of the PrP-lowering divalent siRNA 2439-exNA

4 Reference Standard

Identification:	PrP-lowering divalent siRNA 2439-exNA (shortened to 2439-exNA in this document)
Structure:	Refer to Section 3
Physical Description:	Slightly yellow solution
Purity:	96.6 % (determined by non-denaturing HPLC)
Total Oligo Concentration:	9.9 mg/mL (determined by UV spectrophotometry)
Full-Length Concentration:	9.5634 mg/mL (corrected for purity stated on the CoA)
Supplier:	Argonaut Manufacturing Services (Carlsbad, CA, USA)
Lot No.:	240621005
Manufacturing Date:	24-Jun-2024
Retest Date:	24-Jun-2026

With the exception of a document-based inspection and control of the corresponding Certificate of Analysis (CoA), no further testing on identity and purity of the reference standard was conducted upon receipt.

As the reference standard was supplied by Argonaut Manufacturing Services as a liquid formulation and could be used directly for the preparation of the working solutions (using 0.1 % SDS as diluent), separate aliquots were used for the calibration standards and the Quality Control (QC) samples. For this purpose, the reference standard was previously aliquoted into 100 µL aliquots (Eppendorf Protein LoBind tubes). After aliquoting, the reference standard was renamed to 2439-exNA_240621005_Stock1_04-Sep-2024 in accordance with internal nomenclature. For the preparation of the working solutions for the calibration standards and the Quality Control (QC) samples, the full-length concentration of the reference standard was used.

5 Blank Matrices

Blank matrices used in this method validation are listed in Table 3.

Table 3: Blank Matrices

Blank Matrix (Anticoagulant)	Sex	Receipt Date	Supplier	Lot No.	Expiry Date	Analysis Batch
Beagle dog plasma (anticoagulated with K_2-EDTA)	Male	13-Jan-2023	Biotrend Chemikalien GmbH, Cologne, Germany	BGL142674	31-Dec-2027	AB01 – AB04
	Male			BGL142675		AB01 – AB03
	Male			BGL142676		
	Female			BGL142677		

Blank Matrix (Anticoagulant)	Sex	Receipt Date	Supplier	Lot No.	Expiry Date	Analysis Batch
	Female			BGL142678		
	Female			BGL142679		AB01 – AB04
Beagle dog plasma (hemolyzed, anticoagulated with K_2-EDTA)*1	Mix	08-Jun-2021*2	Biotrend Chemikalien GmbH, Cologne, Germany*2	BGL128811_H _22-Sep-2023	31-May-2026*2	AB01

*1 Hemolyzed plasma was prepared in-house according to KU-RGX-SOP-004 (formerly SOP REG-004) by spiking 2 % (v/v) hemolyzed whole blood (male, K_2-EDTA anticoagulated Beagle dog whole blood supplied by Dunn Labortechnik GmbH, Asbach, Germany, lot: BGL147067, receipt date: 05-Jul-2023, expiry date: 30-Jun-2028) into blank dog plasma.

*2 The data refer to the main component of the hemolyzed plasma, female beagle dog plasma with the lot: BGL128811.

6 PNA Probe

PNA Probe: PNA-BroadInst-01 from HLB Panagene Co., Ltd. (Daejeon, South Korea):

- N-terminus-Atto425 – OO – cdc ttt gtg dgt d – C-terminus, complementary to the antisense strands of 2439-exNA (OO: polyethylene glycol linker; d: 2,6-diaminopurine base modification)
  - Bulk powder was dissolved in 10 % acetonitrile (ACN) and 0.01 % Tween 20
  - Stock solution was diluted with 10 % ACN and 0.01 % Tween 20 to 2 µM (working solution)
- Lot No.: 241164
- Expiry Date: 22-Jul-2027

7 Reference Standard, PNA Probe, and Blank Matrices – Inventory and Disposition

Records of the receipt, distribution and storage of the reference standard, PNA probe, and blank matrices were maintained.

Unused reference standard will be retained for use in subsequent studies for the Sponsor.

Unused PNA probe and blank matrices will be retained for use in subsequent studies.

Expired reference standard will be disposed of unless the shelf life can be extended (new CoA required).

Expired PNA probe and blank matrices will be disposed of or will only be used for testing in non-regulated conditions.

8 METHOD

8.1 Method Summary

This report describes the validation of an AEX-HPLC method with fluorescence detection that allows the sensitive and specific detection of the antisense strands of 2439-exNA, a prion protein (PrP)-lowering divalent siRNA, in dog plasma samples.

The assay is based on the specific hybridization of a 13-mer PNA probe conjugated at the N-terminus with an Atto425 dye that is complementary to the antisense strands of the divalent siRNA 2439-exNA.

The assay is a one-pot approach in which dog plasma lysates are prepared by Proteinase K treatment and the antisense strands of 2439-exNA are hybridized with the PNA in the same step. Subsequently, the samples are analyzed by AEX-HPLC coupled to a fluorescence detector.

In accordance with the MDR, the calibration standards and QC samples were diluted with 0.1 % SDS and the samples (QC-500xULOQ for LTS and dilution integrity) were diluted with 4 M urea. The injected sample volume was 50 µL, resulting in 2.5 µL plasma on column.

Quantification was performed based on an external calibration curve generated from a standard dilution series in dog plasma. Linear calibration curves (weighted 1/X) were calculated from 1.0 ng/mL to 1,000.0 ng/mL.

The detailed procedure is described in the STM, which is included as a supplemental file to this report.

8.2 Applicable Documents

- Axolabs Method Validation Plan for the analysis of 2439-exNA in dog plasma (Axolabs Project: AN402.0-24)
- Method Development Report (MDR) *MDR_2439exNA_DogPlasma* (edition: 02, effective date: 20-Aug-2024)
- Standard Test Method (STM) *STM_Broad_2439-exNA_DogPlasma* (revision 01, effective date: 04-Oct-2024)
- KU-RGL-SOP-002, Revision 03 (formerly SOP REG-102) – Methoden-Validierung im GLP/GCP-Bereich (*Method Validation in GLP/GCP Area*)
- KU-RGL-SOP-009, Revision 03 (formerly SOP REG-109, version 03) – Durchführung des PNA-HPLC-Assay (*Execution of the PNA-HPLC-Assay*)
  - KU-RGL-INS-005, Revision 05 (formerly REG-109-I01) – PNA-HPLC-Assay SST
- KU-RGX-SOP-004, Revision 04 (formerly SOP REG-004, version 04) – Referenzstandards und biologische Matrices (*Reference Standards and Biological Matrices*)

- KU-RGX-FRM-017, Revision 01 (formerly REG-004-F03, version 01) – Herstellung von hemolysiertem Plasma (*Preparation of hemolyzed plasma*)

8.3 Computerized Systems

Critical computerized systems used in this method validation are listed in Table 4.

Table 4: Computerized Systems

System Name	Revision / Version	System Description
Thermo Fisher Ultimate 3000/Shimadzu LC-2050C HPLC System with a Shimadzu RF-20Axs Fluorescence Detector used with Chromeleon HPLC Software	7.2 SR5	HPLC Data collection and export to Excel
Microsoft Excel Spreadsheet: KU-RGL-ESS-002 (formerly REG-102-S02) – Method Validation Data Report	03	Calculates results of calibration standard and QC sample accuracy and precision
Microsoft Excel Spreadsheet: KU-RGL-ESS-005 (formerly REG-102-S05 – Method Validation Protocol PNA Assay - Liquid Matrices - OnePot	01	Used as instruction and as the protocol for the assay execution for liquid matrices using the OnePot method: Calculates all sample preparation steps and the HPLC sequence for an analysis batch
Microsoft Excel Spreadsheet: KU-RGL-ESS-014 (formerly REG-109-S04 – Preparation of HPLC Buffers & Stock Solutions	02	Calculates all buffer preparation steps, including the preparation of the required stock solutions

8.4 HPLC System and Parameters

The HPLC systems used for the AEX-HPLC analyses are listed in Table 5.

Table 5: HPLC Systems

HPLC System	Module	Supplier	Model	Serial No.
HPLC-GLP-7 (Used for AB02 – AB04)	Degasser/Solvent Rack	Thermo Fisher Scientific	Ultimate SR-3000	7294808
	Pump		Ultimate LPG-3400SD	8135975
	Autosampler		Ultimate WPS-3000TSL	8135990
	Column Oven		Ultimate TCC-3000RS	6016238
	Fluorescence Detector	Shimadzu	RF-20AXS	L20505573772
	Communication Unit		CBM-20A	L20235570188
HPLC System	**Module**	**Supplier**	**Model**	**Serial No.**
HPLC-GLP-S-18 (used for AB01)	HPLC Kompakt System	Shimadzu	LC-2050C	L22975950036
	Fluorescence Detector		RF-20AXS	L20505974379

- Eluent Buffer A: 25 mM Tris, pH = 8, 1 mM EDTA, 30 % ACN, 3 M urea
- Eluent Buffer B: 25 mM Tris, pH = 8, 1 mM EDTA, 30 % ACN, 3 M urea, 1 M NaBr
- Eluent Buffer C: 10 % ACN, 4 M NaClO4, 2 M urea, 0.5x Triphosphate Buffer, adjusted to pH 2.5 with phosphoric acid
- Flow Rate: 1 mL/min
- Detection: Excitation @ 436 nm; Emission @ 484 nm; Gain: Medium 16
- Injection of 50 µL of the hybridized sample (corresponds to 2.5 µL matrix on column)
- Column Temperature: 45 °C
- Autosampler Temperature: 20 °C
- HPLC Program: Broad

Table 6: HPLC Gradient Table

Time [min]	% Eluent A	% Eluent B	% Eluent C	Curve
0.00	82	18	0	linear
1.00	82	18	0	linear
5.00	50	50	0	linear
5.10	0	0	100	linear
6.10	0	0	100	linear
6.20	82	18	0	linear
6.70	82	18	0	linear
6.80	0	0	100	linear
7.80	0	0	100	linear
7.90	82	18	0	linear
8.40	82	18	0	linear
8.50	0	0	100	linear
9.50	0	0	100	linear
9.60	82	18	0	linear
13.50	82	18	0	linear

8.5 HPLC Columns

Thermo Fisher Scientific DNAPac PA200 (4 x 250 mm) Column; Serial Number 010629; Lot Number 02306174 (used for AB01)

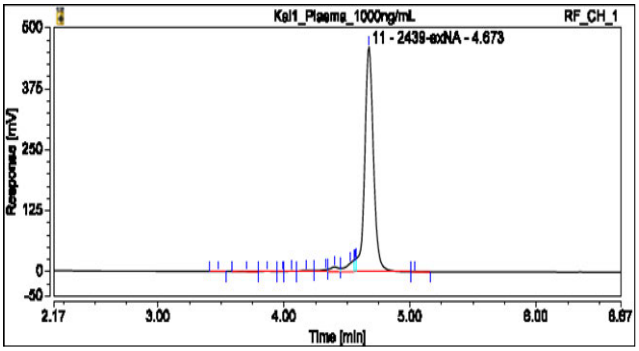
Thermo Fisher Scientific DNAPac PA200 (4 x 250 mm) Column; Serial Number 010628; Lot Number 02306174 (used for AB02 – AB03)

8.6 Data Evaluation

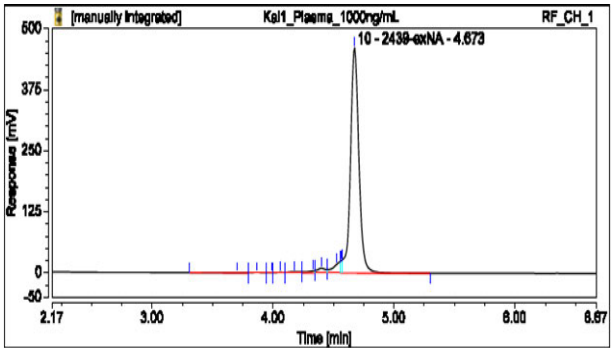
Data acquisition and integration were performed using the software package Chromeleon (Version 7.2) from Thermo Fisher Scientific.

For every injection, the peaks were integrated automatically by the HPLC chromatography software but are adjusted manually if the baseline and peak shoulders are not detected correctly by the HPLC software. When necessary, manual integration was performed by vertical peak splits at minima between two not fully resolved peaks. General rules for manual integration are documented in KU-RGL-SOP-009 (formerly SOP REG-109).

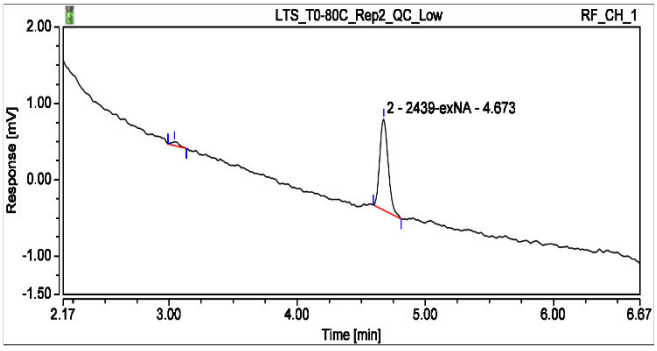
A report format of this file is generated containing values of retention time, peak area, and concentration of 2439-exNA in dog plasma and was saved in PDF format.



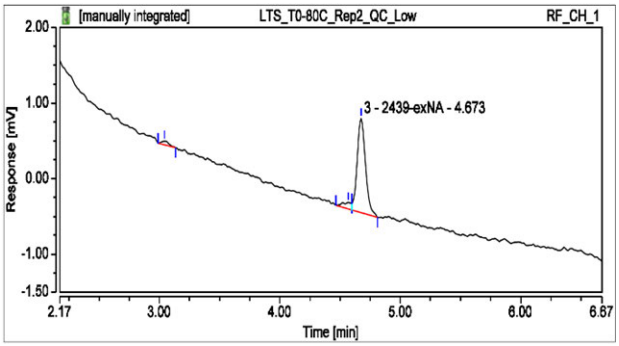
HPLC chromatogram of Cal-level 7 – 1,000.0 ng/mL, sample no. 8 in AB03 **(original)**



HPLC chromatogram of Cal-level 7 – 1,000.0 ng/mL, sample no. 8 in AB03 **(re-integrated)**



HPLC chromatogram of QC-Low LTS sample (T= 0), sample no. 19 in AB03 **(original)**



HPLC chromatogram of QC-Low LTS sample (T= 0), sample no 19 in AB03 **(re-integrated)**

Figure 2: Representative HPLC chromatograms (original and re-integrated)

8.7 Statistical Analysis

Mean values, standard deviations (SD), and coefficients of variation (CV) were calculated using validated Microsoft Excel spreadsheets.

9 Results and Discussion

9.1 System Suitability Test (SST)

The reproducibility of the HPLC system was determined by conducting an SST according to KU-RGL-INS-005 (formerly REG-109-I01) before measuring an analysis batch. Acceptance criteria are described in KU-RGL-INS-005 (formerly REG-109-I01), and the results of the SST run are attached to the documentation of each analysis batch.

A successful SST was completed before each reported analysis batch.

9.2 Calibration Curve

Calibration standards were generated by spiking known concentrations of 2439-exNA into blank dog plasma. The linear range was determined from 1.0 ng/mL to 1,000.0 ng/mL. The calibration levels used were 1.0 ng/mL, 5.0 ng/mL, 50.0 ng/mL, 100.0 ng/mL, 500.0 ng/mL, 1,000.0 ng/mL and one blank sample (dog plasma processed in the absence of 2439-exNA).

The calibration curves were prepared in duplicate for each analysis batch. Blank samples and calibration standards not meeting the acceptance criteria were not used to calculate the calibration curves.

As part of the main validation, in total 6 calibration curves were analyzed in 3 different analysis batches.

9.2.1 Concentration-Response

The concentration-response relation was calculated by linear regression of the mean peak responses of the 6 calibration standards using 1/X weighting.

The calibration curve parameters (slope, y-axis offset, and coefficient of determination) from reported analysis batches are listed in Table 8.

9.2.2 Calibration Curve Accuracy

Acceptance Criteria:

- The back-calculated concentrations of the calibration standards should be within ± 15 % of the nominal concentrations (± 20 % for the LLOQ calibration standard).
- At least 75 % of the calibration standards, with a minimum of 6 calibration standard levels, and at least 50 % of the calibration standards per concentration level must fulfill this criterion.

Results:

- 100.0 % of the back-calculated concentrations for all calibration standards were within ± 15 % (± 20 % for the LLOQ calibration standard) of the nominal concentrations.

Data are listed in Table 9.

Based on these results, the method can be used to determine concentrations of 2439-exNA in dog plasma over the concentration range of 1.0 ng/mL to 1,000.0 ng/mL.

9.2.3 Sensitivity/Lower Limit of Quantification (LLOQ)

The sensitivity of the assay is defined as the lowest 2439-exNA concentration that can be quantified with acceptable accuracy and precision – i.e., the LLOQ.

To confirm the LLOQ, 6 individually prepared samples with a nominal concentration at the LLOQ were analyzed (LLOQ calibration standards).

Acceptance Criteria:

- The analyte response (peak area) in each LLOQ calibration standard should be at least 5 times higher than the response (peak area) at the retention time of the analyte in the blank sample of the same calibration curve.
- The mean back-calculated concentration of the 6 LLOQ calibration standards should be within ± 20 % of the nominal concentration and the coefficient of variation (CV) should not exceed 20 %.

Results:

- No signal at the retention time of 2439-exNA was detected in the blank samples of the calibration curves and hence, the 2439-exNA response (peak area) in each LLOQ calibration standard was at least 5 times higher than the response (peak area) in the blank sample of the same calibration curve.
- The mean back-calculated concentration of the 6 LLOQ calibration standards was 98.3 % of the nominal concentration with a CV of 7.7 %.

Data are listed in Table 9 and Table 10.

The LLOQ for 2439-exNA was confirmed at 1.0 ng/mL in dog plasma.

9.2.4 Upper Limit of Quantification (ULOQ)

The highest calibration standard in the calibration curve defines the ULOQ.

Six (6) individually prepared samples with a nominal concentration at the ULOQ were analyzed (ULOQ calibration standards).

Acceptance Criteria:

The mean back-calculated concentration of the 6 ULOQ calibration standards should be within ± 15 % of the nominal concentration and the CV should not exceed 15 %.

Results:

The mean back-calculated concentration of the 6 ULOQ calibration standards was 99.8 % of the nominal concentration with a CV of 0.9 %.

Data are listed in Table 9.

The ULOQ for 2439-exNA was confirmed at 1,000.0 ng/mL in dog plasma.

9.3 Selectivity

The selectivity of the method was evaluated by comparison of unspiked samples of 7 different blank dog plasma lots (6 non-hemolyzed and non-lipemic matrix lots, 1 hemolyzed matrix lot) with the same dog plasma samples spiked with 2439-exNA at the concentration of the LLOQ (QC-LLOQ).

Acceptance Criteria:

- The analyte response (peak area) in each QC-LLOQ sample should be at least 5 times higher than the response (peak area) at the retention time of the analyte in the blank sample of the same matrix lot.
- The back-calculated concentration each QC-LLOQ sample should be within ± 20 % of the nominal concentration.

Results:

- No signal at the retention time of 2439-exNA was detected in the selectivity blank samples and hence, the 2439-exNA response (peak area) in each LLOQ calibration standard was at least 5 times higher than the response (peak area) in the blank sample of the same matrix lot.
- The back-calculated concentrations of the QC-LLOQ samples ranged from 90.0 % – 100.0 % of the nominal concentration.

Data are listed in Table 11.

Selectivity of the method was confirmed for dog plasma.

9.4 Matrix Effect

The matrix effect was evaluated by analyzing triplicates of QC-Low and QC-High samples, each prepared using 7 different blank dog plasma lots (6 non-hemolyzed and non-lipemic matrix lots, 1 hemolyzed matrix lot).

Acceptance Criteria:

For each evaluated blank dog plasma lot, the mean back-calculated concentration of the QC-Low and QC-High samples (each triplicates) should be within ± 15 % of the nominal concentrations and the CV should not exceed 15 %.

Results:

- For 6 non-hemolyzed/non-lipemic matrix lots (BGL142674 – BGL142679), the mean back-calculated concentrations of the QC samples ranged from 95.6 % – 100.0 % (QC-Low) and 98.2 % – 99.8 % (QC-High) of the nominal concentrations, and CV values ranged from 1.9 % – 3.3 % (QC-Low) and 0.0 % – 0.5 % (QC-High).
- For the hemolyzed matrix lot (BGL128811_H_22-Sep-2023), the mean back-calculated concentrations of the QC samples were 100.0 % (QC-Low) and 99.4 % (QC-High) of the nominal concentrations, and CV values were 0.0 % % (QC-Low) and 0.3 % % (QC-High).

Data are listed in Table 12.

Based on these results, there is no matrix effect for the assay regarding the analysis of 2439-exNA in dog plasma.

9.5 Carry-Over

Blank samples were injected after the ULOQ calibration standards to assess carry-over.

Acceptance Criteria:

The analyte response (peak area) in each LLOQ calibration standard should be at least 5 times higher than the analyte response (peak area) in the carry-over blank sample following the ULOQ calibration standard of the same calibration curve.

Results:

- No signal for 2439-exNA was detected in the carry-over blank samples and hence, the 2439-exNA response (peak area) in each LLOQ calibration standard was at least 5 times higher than the response (peak area) in the carry-over blank sample following the ULOQ calibration standard of the same calibration curve.

Data are listed in Table 13.

No carry-over was observed for 2439-exNA in dog plasma. Therefore, no special precautions after the analysis of samples with a high concentration of 2439-exNA have to be implemented.

9.6 Quality Control (QC) Samples

Analysis of assay accuracy and precision, dilution integrity, and stability was conducted by analysis of QC samples of defined concentrations.

QC samples were prepared by spiking known amounts of 2439-exNA into dog plasma.

QC samples were prepared at 4 concentration levels which cover the calibration curve range:

- at the LLOQ (QC-LLOQ): 1.0 ng/mL
- within 3 times above the LLOQ (QC-Low): 3.0 ng/mL
- at 30 – 50 % of the calibration curve range (QC-Mid): 450.0 ng/mL
- at least at 75 % of the upper calibration curve range (QC-High): 900.0 ng/mL

9.7 Accuracy

The accuracy of the analytical method describes the closeness of the determined concentration obtained by the method to the nominal concentration of 2439-exNA (expressed in percentage).

The analysis of assay accuracy was conducted using the QC samples at the 4 concentration levels that were measured as 6 individually prepared replicates per analytical run.

The analysis of assay accuracy was divided into within-run and between-run accuracy. The within-run accuracy was repeated 3 times, prepared by 2 technicians on different days, and analyzed on 2 independent HPLC systems. Thus, 3 independent data sets for the between-run accuracy were generated.

The concentration of 2439-exNA was calculated against the external calibration curve, generated in duplicate for each data set.

9.7.1 Within-Run Accuracy

Acceptance Criteria:

The mean back-calculated concentration of the QC samples at each of the 4 concentration levels and in each of the 3 data sets should be within ± 15 % of the nominal concentrations (± 20 % for the QC-LLOQ).

Results:

The mean back-calculated concentrations of the QC samples at each of the 4 concentration levels in the 3 data sets ranged from 95.0 % – 98.3 % (QC-LLOQ), 97.2 % – 98.9 % (QC-Low), 97.7 % – 100.8 % (QC-Mid) and 98.3 % – 99.6 % (QC-High) of the nominal concentrations.

Data are listed in Table 14.

9.7.2 Between-Run Accuracy

Acceptance Criteria:

The mean back-calculated concentrations of all QC samples at each of the 4 concentration levels of all 3 data sets should be within ± 15 % of the nominal concentrations (± 20 % for the QC-LLOQ).

Results:

The mean back-calculated concentrations of all QC samples at each of 4 concentration levels of all 3 data sets were 97.2 % (QC-LLOQ), 98.0 % (QC-Low), 99.7 % (QC-Mid), and 99.0 % (QC-High) of the nominal concentrations.

Data are listed in Table 14.

Based on these results, the assay for 2439-exNA in dog plasma is considered accurate.

9.8 Precision

The precision of the analytical method describes the closeness of repeated individual measures of the 2439-exNA concentration, expressed as coefficient of variation (CV).

The analysis of assay precision was conducted using the QC samples at the 4 concentration levels. All concentration levels were measured as 6 individually prepared replicates per analytical run.

As for accuracy, precision was demonstrated within a single analytical run and between different analytical runs. The same analytical runs and data were used for the demonstration of accuracy.

The concentration of 2439-exNA was calculated against the external calibration curve, generated in duplicate for each data set.

9.8.1 Within-Run Precision

Acceptance Criteria:

The within-run CV of the QC samples at each of the 4 concentrations levels and in each of the 3 data sets should not exceed 15 % (20 % for the QC-LLOQ).

Results:

The within-run CV of the QC samples at each of the 4 concentration levels in the 3 data sets ranged from 4.2 % – 7.7 % (QC-LLOQ), 1.8 % – 2.8 % (QC-Low), 0.4 % – 0.6 % (QC-Mid) and 0.2 % – 0.4 % (QC-High).

Data are listed in Table 14.

9.8.2 Between-Run Precision

Between-run precision was determined with an identical setup as for within-run precision but included the assessment of QC samples from 3 independent data sets prepared by 2 technicians on 3 different days on 2 independent HPLC systems.

Acceptance Criteria:

The between-run CV of all QC samples at each of the 4 concentration levels of all 3 data sets should not exceed 15 % (20 % for the QC-LLOQ).

Results:

- The between-run CV of all QC samples at each of the 4 concentration levels of all 3 data sets were 5.9 % (QC-LLOQ), 2.4 % (QC-Low), 1.5 % (QC-Mid), and 0.7 % (QC-High).

Data are listed in Table 14.

Based on these results, the assay for 2439-exNA in dog plasma is considered precise.

9.9 Dilution Integrity

Dilution integrity was evaluated using QC samples prepared at the concentration level 500-fold (500,000.0 ng/mL) above the ULOQ (6 replicates for each dilution factor).

The 500,000.0 ng/mL QC samples (QC-500xULOQ) were analyzed and quantified after a 1:1,000 dilution.

The QC samples were diluted with 4 M urea. During the method development, the suitability of 4 M Urea as a diluent for study samples was shown. Blank dog plasma was not used for sample dilution since dog plasma is a rare matrix. In line with the 3R principles (Reduce, Refine, Replace) it is not ethically justifiable to use the large quantities of blank dog plasma for sample dilution. In terms of animal welfare, 4 M Urea is therefore a good alternative to replace biological matrix.

Acceptance Criteria

For each evaluated dilution factor and concentration, the mean back-calculated concentration of the QC samples should be within ± 15 % of the nominal concentration and the CV should not exceed 15 %.

Results:

- The mean back-calculated concentration was 95.2 % for the 500,000.0 ng/mL QC samples (1:1,000 dilution).
- The CV was 0.7 % for the 500,000.0 ng/mL QC samples (1:1,000 dilution).

Data are listed in Table 15.

Dilution of samples up to 1:1,000 up to a concentration of 500,000.0 ng/mL met the acceptance criteria for precision and accuracy. As the 1:1,000 dilution integrity testing fulfilled the acceptance criteria, no additional dilutions (e.g. 1:500) were tested.

9.10 Stability

The stability of 2439-exNA was analyzed under several conditions relevant to the assay setup. 2439-exNA was spiked at 3 concentration levels into dog plasma samples and analyzed in triplicate. The concentration levels were identical to the QC-Low, QC-High and QC-500xULOQ samples, with concentrations of 3.0 ng/mL (QC-Low), 900.0 ng/mL (QC-High) and 500,000.0 ng/mL (QC-500xULOQ). The QC-500xULOQ samples were evaluated after 1:1,000 dilution (using 4 M urea) to mimic potential samples above the validated ULOQ. Calibration curves were freshly prepared for each analytical run.

Following the initial failed long-term stability measurements, the preparation of the working solutions was adjusted. Deviating from the procedure described in the MDR, the working solutions used for the preparation of the stability samples (LTS, FTS and STS) were prepared in glass vials (instead of Eppendorf Protein LoBind tubes) and 0.1 % SDS (instead of dog plasma) was used as diluent for the working solutions to avoid non-specific binding of the analyte to the sample vial. Aliquoting of the stability samples and subsequent storage was performed in polypropylene tubes (Eppendorf Protein LoBind tubes).

9.10.1 Long-Term Stability

Aliquots of 2439-exNA samples in dog plasma at the 3 QC concentration levels were stored at -80 °C ± 20 °C, and were and will be analyzed at up to 3 time points (T= 0, 3, and 12 months) in triplicate.

Acceptance Criteria:

For each evaluated condition, the mean back-calculated concentration of the QC-Low, QC-High and QC-500xULOQ samples (each triplicates) should be within ± 15 % of the nominal concentrations.

Results:

At the time the Method Validation Report was prepared, results of the first time point (T= 0) were available.

- At T= 0 in AB01, the mean back-calculated concentrations were 86.7 % (QC-Low), 90.1 % (QC-High) and 83.6 % (QC-500xULOQ) of the nominal concentrations.

As long-term stability testing at the initial analysis missed acceptance criteria at QC-500xULOQ concentration, analysis was repeated for this concentration, to either confirm or disprove the initial result.

- At T= 0 in AB02 (first repetition), the mean back-calculated concentrations 82.7 % (QC-500xULOQ) of the nominal concentration.

As acceptance criteria for accuracy at QC-500xULOQ concentration were repeatedly missed, the preparation of the reference standard working solutions used for the stability samples was adapted (see Section 9.10).

- At T= 0 in AB03, the mean back-calculated concentrations were 88.9 % (QC-Low), 93.8 % (QC-High) and 87.7 % (QC-500xULOQ) of the nominal concentrations.

Data are listed in Table 16.

Long-term stability of 2439-exNA in dog plasma will be further monitored for up to 12 months. The final long-term stability data set will be added as an addendum to the Method Validation Report at the end of the study.

9.10.2 Freeze-Thaw and Short-Term Stability

Freeze-thaw and short-term stability of 2439-exNA in dog plasma were evaluated in a combined test.

Aliquots of 2439-exNA samples in dog plasma at the 3 QC concentration levels were stored at -80 °C ± 20 °C for at least 12 hours.

Thawing of the QC samples from -80 °C ± 20 °C was done on ice. After thawing, the QC samples were stored on ice for 4 hours before further processing, covering short-term stability requirements.

Subsequently, the QC samples were refrozen for at least 12 hours at -80 °C ± 20 °C to start the next freeze-thaw cycle and to repeat the process described above.

After 5 freeze-thaw cycles, the QC samples were processed and analyzed in triplicate according to the assay conditions.

Acceptance Criteria:

The mean back-calculated concentration of the QC-Low, QC-High and QC-500xULOQ samples (each triplicates) should be within ± 15 % of the nominal concentrations.

Results:

- After 5 freeze/thaw cycles with 4 hours on ice, the mean back-calculated concentrations were 93.3 % (QC-Low), 94.2 % (QC-High) and 86.8 % (QC-500xULOQ) of the nominal concentrations.

Data are listed in Table 17.

At the evaluated concentrations, 2439-exNA is stable in dog plasma when thawed up to 5 times on ice during sample analysis and for up to 4 hours during sample storage on ice.

9.10.3 Bench-Top Stability/Robustness

Since the assay is set up as a one-pot method that does not have separate lysis and hybridization steps, the bench-top stability was covered by the short-term stability (see Section 9.10.2).

9.10.4 Stability of the Processed Sample/On-Instrument Stability (Autosampler)

Stability of the processed sample of 2439-exNA in dog plasma was assessed with 4 QC sets of QC-Low and QC-High triplicates that were stored on the HPLC autosampler at 20 °C and were analyzed at different time points, covering 77 hours between the analysis of the first and the last QC set.

Acceptance Criteria:

- For each evaluated time point, the mean back-calculated concentration of the QC-Low and QC-High samples should be within ± 15 % of the nominal concentrations.
- The QC set of last time point, which fulfills the acceptance criteria, defines processed sample stability.

Results:

- After time point 1, the mean back-calculated concentrations were 93.3 % (QC-Low) and 99.4 % (QC-High) of the nominal concentrations.
- After time point 2 (9 hours), the mean back-calculated concentrations were 95.6 % (QC-Low) and 100.3 % (QC-High) of the nominal concentrations.
- After time point 3 (47 hours), the mean back-calculated concentrations were 96.7 % (QC-Low) and 102.4 % (QC-High) of the nominal concentrations.

In AB03, sample no. 11 (first replicate of QC-Low) could be designated as an outlier at a probability of at least 95 % according to W. Gottwald, Statistik für Anwender, Wiley-VCH Verlag and was therefore excluded from the calculation of mean, SD and CV.

- After time point 4 (77 hours), the mean back-calculated concentrations were 97.8 % (QC-Low) and 103.1 % (QC-High) of the nominal concentrations.

Data are listed in Table 18.

At the evaluated concentrations, 2439-exNA in processed dog plasma samples is stable during storage on the HPLC autosampler at 20 ± 3 °C for up to 77 hours.

If a longer period of sample storage on the autosampler should be necessary, the last QC sample set and calibration curve will verify the validity of the analysis.

9.10.5 Stability of Stock Solution and Working Solutions

9.10.5.1 Short-Term Stability of Stock Solution and Working Solutions

Aliquots of the reference standard stock solution of 2439-exNA with a concentration of approximately 9.5634 mg/mL in 1x DPBS (total oligonucleotide concentration corrected for the purity stated in the CoA), and the highest and the lowest concentration of the working solutions (working solution LLOQ/ULOQ) of 2439-exNA were stored at room temperature for 4 hours and analyzed in triplicate. The reference standard stock solution was diluted to QC-High concentration before analysis.

The samples were analyzed by comparing the instrument's response with a freshly thawed reference standard stock solution and freshly prepared working solutions of the same concentrations.

Acceptance Criteria:

The mean back-calculated concentration of the QC-High samples (reference standard stock solution) and the working solution samples (each triplicates) should be within ± 15 % of the nominal concentrations (± 20 % for the working solution LLOQ).

Results:

After 4 hours at room temperature, the mean back-calculated concentrations were 96.7 % (working solution LLOQ), 95.5 % (working solution ULOQ), and 99.1 % (stock solution) of the nominal concentrations.

Data are listed in Table 19.

At the evaluated concentrations, 2439-exNA stock and working solutions covering the complete assay range are stable during storage for up to 4 hours at room temperature.

9.10.5.2 Long-Term Stability of Stock Solution

Long-term stability of the reference standard stock solution was not performed, as stability data will be provided by the Sponsor.

9.11 Other Validation Runs

For other validation runs, the calibration curve accuracy were assessed with 6 calibration standards analyzed in duplicate and with QC samples at 3 concentration levels (QC-Low, QC-Mid, QC-High) analyzed at least in duplicates (or at least 5 % of the number of samples, whichever was higher).

9.11.1 Calibration Curve

The calibration curves in other validation runs were prepared with 6 calibration standards in duplicate using the same setup and calculation as described in Section 9.2.

9.11.1.1 Concentration-Response

The concentration-response relation in other validation runs was calculated by linear regression of the mean peak responses of the 6 calibration standards using 1/X weighting.

The calibration curve parameters (slope, y-axis offset, and correlation coefficient) from other validation runs are reported in Table 20.

9.11.1.2 Calibration Curve Accuracy

Acceptance Criteria:

In each other validation run, the back-calculated concentrations of the calibration standards should be within ± 15 % of the nominal concentrations (± 20 % for the LLOQ calibration standard).

At least 75 % of all calibration standards, with a minimum of 6 calibration standard levels, and at least 50 % of the calibration standards per concentration level must fulfill this criterion.

Results:

- In AB04, 100 % of the back-calculated concentrations for all calibration standards were within ± 15 % (± 20 % for the LLOQ calibration standard) of the nominal concentrations.

Data are listed in Table 21.

Acceptance criteria for the calibration standards were fulfilled for other validation runs.

9.11.2 QC Samples

In other validation runs, QC samples were prepared as described in Section 9.6 at 3 concentration levels:

- at 3 times above the LLOQ (QC-Low): 3.0 ng/mL
- within 30 – 50 % of the calibration curve range (QC-Mid): 450.0 ng/mL
- at least at 75 % of the upper calibration curve range (QC-High): 900.0 ng/mL

9.11.2.1 QC Samples Accuracy

Acceptance Criteria:

In each other validation run, the back-calculated concentrations of at least 2/3 of the QC samples and of at least 50 % of the QC samples at each concentration level should be within ± 15 % of the nominal concentrations.

Results:

- In AB04, 6 out of 6 QC samples (100 %) were within ± 15 % of the nominal concentrations.

Data are listed in Table 22.

Acceptance criteria for the QC samples were fulfilled for other validation runs.

9.11.3 Selectivity and Carry-Over

9.11.3.1 Selectivity

For the selectivity test in other validation runs, the analyte response (peak area) of the blank samples from the calibration curves was evaluated for interference in comparison to the LLOQ calibration standards.

Acceptance Criteria:

The analyte response (peak area) in each LLOQ calibration standard should be at least 5 times higher than the response (peak area) at the retention time of the analyte in the blank sample of the same calibration curve.

Results:

- No signal at the retention time of 2439-exNA was detected in the blank samples of the calibration curves in other validation runs and hence, the 2439-exNA response (peak area) in each LLOQ calibration standard was at least 5 times higher than the response (peak area) in the blank sample of the same calibration curve.

Data are listed in Table 23.

The acceptance criteria for selectivity were fulfilled for other validation runs.

9.11.3.2 Carry-Over

Blank samples were injected after the ULOQ calibration standard to assess carry-over in other validation runs.

Acceptance Criteria:

The analyte response (peak area) in each LLOQ calibration standard should be at least 5 times higher than the analyte response (peak area) in the carry-over blank sample following the ULOQ calibration standard of the same calibration curve.

Results:

- No signal for 2439-exNA was detected in the carry-over blank samples and hence, the 2439-exNA response (peak area) in each LLOQ calibration standard was at least 5 times higher than the response (peak area) in the carry-over blank sample following the ULOQ calibration standard of the same calibration curve.

Data are listed in Table 23.

Acceptance criteria for carry-over were fulfilled for other validation runs.

10 Deviations

The bioanalytical method validation for the analysis of 2439-exNA in dog plasma was conducted according to the Method Validation Plan and to applicable SOPs without any deviation.

11 Conclusions

The bioanalytical method for the analysis of 2439-exNA in dog plasma was successfully validated and is described in the *STM_Broad_2439-exNA_DogPlasma* (Revision 01, effective date: 04-Oct-2024). All validation tests fulfilled the acceptance criteria described in the Method Validation Plan.

The method can be used within a linear range from 1.0 ng/mL to 1,000.0 ng/mL of 2439-exNA in dog plasma, applying a 1/X weighted linear regression model.

2439-exNA is stable in dog plasma during up to 5 freeze/thaw cycles. The evaluation of long-term stability of 2439-exNA in dog plasma is still ongoing, and the results will be added as an addendum to this Method Validation Report after the finalization of the long-term stability testing.

12 List of Records to be Retained

All original data and documents generated during this method validation will be archived at Axolabs GmbH or at an Axolabs-employed GLP-compliant archive facility under appropriate conditions in accordance with German and EU legislation, and KU-RGL-SOP-005 (formerly SOP REG-105).

The following records will be retained:

- Final signed Method Validation Plan
- Documentation of deviations and corrections
- Critical communication, as appropriate
- Raw data:
  - HPLC file(s) with HPLC chromatograms, audit trails, and HPLC program
  - Assay protocols
  - System Suitability Test reports
  - CoA or equivalent of PNA probe, blank matrices, and reference standard
  - Other raw data (if applicable)
- QC and QA records of document and data reviews/audits
- Final signed Method Validation Report and all addenda

A backup file of all analytical HPLC data, together with an electronic PDF report of all analytical runs, and all electronic files generated during the method validation (reports, Excel-Files) are stored in electronic form.

13 Data Tables

Table 7: Analytical Batches for the Analysis of 2439-exNA in Dog Plasma

Run ID	Date of Analysis	Description	Status	Comments
AN402.0-24_01_AB01	06-Sep-2024	1st main validation batch	Passed	QC-500xULOQ (LTS, T= 0) failed acceptance criterion
AN402.0-24_01_AB02	10-Sep-2024	2nd main validation batch, repetition of T= 0 of LTS (QC-500xULOQ)	Passed	QC-500xULOQ (LTS, T= 0) failed acceptance criterion
AN402.0-24_01_AB03	12-Sep-2024	3rd main validation batch, restart of LTS measurement	Passed	Preparation of a new set of stability samples, sample no. 11 (QC-Low of PSS) was excluded from the statistical calculation after performing an outlier test
AN402.0-24_01_AB04	18-Sep-2024	Other validation run, measurement of FTS (5 cycles)	Passed	n.a.

Table 8: Calibration Curve Parameters for the Analysis of 2439-exNA in Dog Plasma

Equation for the calibration curve using a linear regression model weighted 1/X: y = m*x + b					
Run ID	**Date of Analysis**	**Sample No.**	**Slope [m]**	**Y-Offset [b]**	**R-Squared**
AB01	06-Sep-2024	5 – 10, 84 – 89	0.02524	-0.00159	0.99999
AB02	10-Sep-2024	3 – 8, 58 – 63	0.03830	-0.00010	0.99992
AB03	12-Sep-2024	3 – 8, 66 – 71	0.03764	-0.00115	0.99994

Table 9: Results of the Calibration Standards for the Analysis of 2439-exNA in Dog Plasma

		Cal-Level-2 (LLOQ)		Cal-Level-3		Cal-Level-4		Cal-Level-5		Cal-Level-6		Cal-Level-7 (ULOQ)	
		1.0 ng/mL		5.0 ng/mL		50.0 ng/mL		100.0 ng/mL		500.0 ng/mL		1,000.0 ng/mL	
Run ID	**Sample No.**	**Calculated 2439-exNA Conc. ng/mL**	**Acc. Criterion ± 20 % of Nominal**	**Calculated 2439-exNA Conc. ng/mL**	**Acc. Criterion ± 15 % of Nominal**	**Calculated 2439-exNA Conc. ng/mL**	**Acc. Criterion ± 15 % of Nominal**	**Calculated 2439-exNA Conc. ng/mL**	**Acc. Criterion ± 15 % of Nominal**	**Calculated 2439-exNA Conc. ng/mL**	**Acc. Criterion ± 15 % of Nominal**	**Calculated 2439-exNA Conc. ng/mL**	**Acc. Criterion ± 15 % of Nominal**
AB01	5 – 10	1.0	100.0 %	5.0	100.0 %	50.1	100.2 %	100.6	100.6 %	500.1	100.0 %	1,002.7	100.3 %
AB01	84 – 89	1.0	100.0 %	5.2	104.0 %	50.3	100.6 %	100.8	100.8 %	499.1	99.8 %	996.3	99.6 %
AB02	3 – 8	0.9	90.0 %	4.8	96.0 %	49.8	99.6 %	99.5	99.5 %	493.9	98.8 %	988.3	98.8 %
AB02	58 – 63	1.1	110.0 %	5.0	100.0 %	51.0	102.0 %	103.1	103.1 %	507.0	101.4 %	1,007.6	100.8 %
AB03	3 – 8	1.0	100.0 %	5.0	100.0 %	49.7	99.4 %	100.0	100.0 %	496.2	99.2 %	987.8	98.8 %
AB03	66 – 71	0.9	90.0 %	5.0	100.0 %	51.3	102.6 %	102.8	102.8 %	505.8	101.2 %	1,006.4	100.6 %
	Mean	0.98	98.3 %	5.00	100.0 %	50.37	100.7 %	101.13	101.1 %	500.35	100.1 %	998.18	99.8 %
	SD (±)	0.08		0.13		0.65		1.48		5.18		8.78	
	CV (%)	7.7 %		2.5 %		1.3 %		1.5 %		1.0 %		0.9 %	
	N	6		6		6		6		6		6	

Table 10: Sensitivity/Lower Limit of Quantification (LLOQ) for the Analysis of 2439-exNA in Dog Plasma

		Cal-Level-1 (Blank Sample)		Cal-Level-2 (LLOQ)	
		0.0 ng/mL		1.0 ng/mL	
Run ID	**Sample No.**	**Peak Area [mV*min]**	**Acc. Criterion ≤ 20 % of LLOQ**	**Peak Area [mV*min]**	**Calculated 2439-exNA Conc. ng/mL**
AB01	4, 5	0.0000	0.0 %	0.0226	1.0
AB01	83, 84	0.0000	0.0 %	0.0233	1.0
AB02	2, 3	0.0000	0.0 %	0.0356	0.9
AB02	57, 58	0.0000	0.0 %	0.0408	1.1
AB03	2, 3	0.0000	0.0 %	0.0379	1.0
AB03	65, 66	0.0000	0.0 %	0.0330	0.9

Table 11: Selectivity for the Analysis of 2439-exNA in Dog Plasma

					Blank Sample for Selectivity		QC-LLOQ		
					0.0 ng/mL		1.0 ng/mL		
Run ID	**Date of Analysis**	**Sex**	**Blank Matrix Lot No.**	**Sample No.**	**Peak Area [mV*min]**	**Acc. Criterion ≤ 20 % of LLOQ**	**Peak Area [mV*min]**	**Calculated 2439-exNA Conc. ng/mL**	**Acc. Criterion ± 20 % of Nominal**
AB01	06-Sep-2024	Male	BGL142674	23, 31	0.0000	0.0 %	0.0230	1.0	100.0 %
AB01	06-Sep-2024	Male	BGL142675	24, 32	0.0000	0.0 %	0.0233	1.0	100.0 %
AB01	06-Sep-2024	Male	BGL142676	25, 33	0.0000	0.0 %	0.0231	1.0	100.0 %
AB01	06-Sep-2024	Female	BGL142677	26, 34	0.0000	0.0 %	0.0214	0.9	90.0 %
AB01	06-Sep-2024	Female	BGL142678	27, 35	0.0000	0.0 %	0.0230	1.0	100.0 %
AB01	06-Sep-2024	Female	BGL142679	28, 36	0.0000	0.0 %	0.0229	1.0	100.0 %
AB01	06-Sep-2024	Mix	BGL128811_H_22-Sep-2023 (Hemolyzed)	29, 37	0.0000	0.0 %	0.0242	1.0	100.0 %

Table 12: Matrix Effect for the Analysis of 2439-exNA in Dog Plasma

			QC-Low		QC-High	
			3.0 ng/mL		900.0 ng/mL	
Run ID	**Blank Matrix Lot No.**	**Sample No.**	**Calculated 2439-exNA Conc. ng/mL**	**Acc. Criterion ± 15 % of Nominal**	**Calculated 2439-exNA Conc. ng/mL**	**Acc. Criterion ± 15 % of Nominal**
AB01	BGL142674	39, 56	2.9	96.7 %	886.0	98.4 %
		40, 57	3.0	100.0 %	881.1	97.9 %
		41, 58	2.9	96.7 %	885.0	98.3 %
		Mean	2.93	97.8 %	884.03	98.2 %
		SD (±)	0.06		2.59	
		CV (%)	2.0 %		0.3 %	
		N	3		3	
AB01	BGL142675	42, 59	3.1	103.3 %	889.1	98.8 %
		43, 60	3.0	100.0 %	884.9	98.3 %
		44, 61	2.9	96.7 %	879.9	97.8 %
		Mean	3.00	100.0 %	884.63	98.3 %
		SD (±)	0.10		4.61	
		CV (%)	3.3 %		0.5 %	
		N	3		3	

Table 12: Matrix Effect for the Analysis of 2439-exNA in Dog Plasma (Continued)

			QC-Low		QC-High	
			3.0 ng/mL		900.0 ng/mL	
Run ID	**Blank Matrix Lot No.**	**Sample No.**	**Calculated 2439-exNA Conc. ng/mL**	**Acc. Criterion ± 15 % of Nominal**	**Calculated 2439-exNA Conc. ng/mL**	**Acc. Criterion ± 15 % of Nominal**
AB02	BGL142676	25, 39	3.0	100.0 %	892.6	99.2 %
		26, 40	2.9	96.7 %	892.1	99.1 %
		27, 41	2.9	96.7 %	892.7	99.2 %
		Mean	2.93	97.8 %	892.47	99.2 %
		SD (±)	0.06		0.32	
		CV (%)	2.0 %		0.0 %	
		N	3		3	
AB02	BGL142677	28, 42	3.0	100.0 %	893.2	99.2 %
		29, 43	2.9	96.7 %	890.3	98.9 %
		30, 44	2.9	96.7 %	896.8	99.6 %
		Mean	2.93	97.8 %	893.43	99.3 %
		SD (±)	0.06		3.26	
		CV (%)	2.0 %		0.4 %	
		N	3		3	

Table 12: Matrix Effect for the Analysis of 2439-exNA in Dog Plasma (Continued)

			QC-Low		QC-High	
			3.0 ng/mL		900.0 ng/mL	
Run ID	**Blank Matrix Lot No.**	**Sample No.**	**Calculated 2439-exNA Conc. ng/mL**	**Acc. Criterion ± 15 % of Nominal**	**Calculated 2439-exNA Conc. ng/mL**	**Acc. Criterion ± 15 % of Nominal**
AB03	BGL142678	35, 49	2.9	96.7 %	896.4	99.6 %
		36, 50	3.0	100.0 %	900.8	100.1 %
		37, 51	3.0	100.0 %	897.2	99.7 %
		Mean	2.97	98.9 %	898.13	99.8 %
		SD (±)	0.06		2.34	
		CV (%)	1.9 %		0.3 %	
		N	3		3	
AB03	BGL142679	38, 52	2.9	96.7 %	890.5	98.9 %
		39, 53	2.9	96.7 %	897.9	99.8 %
		40, 54	2.8	93.3 %	895.8	99.5 %
		Mean	2.87	95.6 %	894.73	99.4 %
		SD (±)	0.06		3.81	
		CV (%)	2.0 %		0.4 %	
		N	3		3	

Table 12: Matrix Effect for the Analysis of 2439-exNA in Dog Plasma (Continued)

			QC-Low		QC-High	
			3.0 ng/mL		900.0 ng/mL	
Run ID	**Blank Matrix Lot No.**	**Sample No.**	**Calculated 2439-exNA Conc. ng/mL**	**Acc. Criterion ± 15 % of Nominal**	**Calculated 2439-exNA Conc. ng/mL**	**Acc. Criterion ± 15 % of Nominal**
AB01	BGL128811_H_22-Sep-2023 (Hemolyzed)	45, 62	3.0	100.0 %	896.7	99.6 %
		46, 63	3.0	100.0 %	892.1	99.1 %
		47, 64	3.0	100.0 %	893.8	99.3 %
		Mean	3.00	100.0 %	894.20	99.4 %
		SD (±)	0.00		2.33	
		CV (%)	0.0 %		0.3 %	
		N	3		3	

Table 13: Carry-Over for the Analysis of 2439-exNA in Dog Plasma

		Cal-Level-2 (LLOQ)			Cal-Level-7 (ULOQ)			Blank Sample for Carry-Over		
		1.0 ng/mL			1,000.0 ng/mL			0.0 ng/mL		
Run ID	**Sample No.**	**Peak Area [mV*min]**	**Calculated 2439-exNA Conc. ng/mL**	**Acc. Criterion ± 20 % of Nominal**	**Peak Area [mv*min]**	**Calculated 2439-exNA Conc. ng/mL**	**Acc. Criterion ± 15 % of Nominal**	**Peak Area [mV*min]**	**Calculated 2439-exNA Conc. ng/mL**	**Acc. Criterion ≤ 20 % of LLOQ**
AB01	5, 10, 11	0.0226	1.0	100.0 %	25.3018	1,002.7	100.3 %	0.0000	BQL	0.0 %
AB01	84, 89, 90	0.0233	1.0	100.0 %	25.1407	996.3	99.6 %	0.0000	BQL	0.0 %
AB02	3, 8, 9	0.0356	0.9	90.0 %	37.8503	988.3	98.8 %	0.0000	BQL	0.0 %
AB02	58, 63, 64	0.0408	1.1	110.0 %	38.5893	1,007.6	100.8 %	0.0000	BQL	0.0 %
AB03	3, 8, 9	0.0379	1.0	100.0 %	37.1806	987.8	98.8 %	0.0000	BQL	0.0 %
AB03	66, 71, 72	0.0330	0.9	90.0 %	37.8792	1,006.4	100.6 %	0.0000	BQL	0.0 %

Table 14: Results of the QC Samples (Accuracy and Precision) for the Analysis of 2439-exNA in Dog Plasma

					QC-LLOQ		QC-Low		QC-Mid		QC-High	
					1.0 ng/mL		3.0 ng/mL		450.0 ng/mL		900.0 ng/mL	
Run ID	**Date of Analysis**	**User**	**HPLC System**	**Sample No.**	**Calculated 2439-exNA Conc. ng/mL**	**Acc. Criterion ± 20 % of Nominal**	**Calculated 2439-exNA Conc. ng/mL**	**Acc. Criterion ± 15 % of Nominal**	**Calculated 2439-exNA Conc. ng/mL**	**Acc. Criterion ± 15 % of Nominal**	**Calculated 2439-exNA Conc. ng/mL**	**Acc. Criterion ±15 % of Nominal**
AB01	06-Sep-2024	JSt	HPLC-GLP-S-18	31, 39, 49, 56	1.0	100.0 %	2.9	96.7 %	443.5	98.6 %	886.0	98.4 %
				32, 40, 50, 57	1.0	100.0 %	3.0	100.0 %	440.0	97.8 %	881.1	97.9 %
				33, 41, 51, 58	1.0	100.0 %	2.9	96.7 %	439.0	97.6 %	885.0	98.3 %
				34, 42, 52, 59	0.9	90.0 %	3.1	103.3 %	439.0	97.6 %	889.1	98.8 %
				35, 43, 53, 60	1.0	100.0 %	3.0	100.0 %	438.0	97.3 %	884.9	98.3 %
				36, 44, 54, 61	1.0	100.0 %	2.9	96.7 %	438.0	97.3 %	879.9	97.8 %
Within-Run Accuracy/Precision I				**Mean**	0.98	98.3 %	2.97	98.9 %	439.58	97.7 %	884.33	98.3 %
				SD (±)	0.04		0.08		2.06		3.36	
				CV (%)	4.2 %		2.8 %		0.5 %		0.4 %	
				N	6		6		6		6	
AB02	10-Sep-2024	YE	HPLC-GLP-7	18, 25, 32, 39	1.0	100.0 %	3.0	100.0 %	452.0	100.4 %	892.6	99.2 %
				19, 26, 33, 40	1.1	110.0 %	2.9	96.7 %	452.3	100.5 %	892.1	99.1 %
				20, 27, 34, 41	1.0	100.0 %	2.9	96.7 %	456.0	101.3 %	892.7	99.2 %
				21, 28, 35, 42	1.0	100.0 %	3.0	100.0 %	454.0	100.9 %	893.2	99.2 %
				22, 29, 36, 43	0.9	90.0 %	2.9	96.7 %	451.7	100.4 %	890.3	98.9 %
				23, 30, 37, 44	0.9	90.0 %	2.9	96.7 %	454.5	101.0 %	896.8	99.6 %
Within-Run Accuracy/Precision II				**Mean**	0.98	98.3 %	2.93	97.8 %	453.42	100.8 %	892.95	99.2 %
				SD (±)	0.08		0.05		1.70		2.14	
				CV (%)	7.7 %		1.8 %		0.4 %		0.2 %	
				N	6		6		6		6	

Table 14: Results of the QC Samples (Accuracy and Precision) for the Analysis of 2439-exNA in Dog Plasma (Continued)

					QC-LLOQ		QC-Low		QC-Mid		QC-High	
					1.0 ng/mL		3.0 ng/mL		450.0 ng/mL		900.0 ng/mL	
Run ID	**Date of Analysis**	**User**	**HPLC System**	**Sample No.**	**Calculated 2439-exNA Conc. ng/mL**	**Acc. Criterion ± 20 % of Nominal**	**Calculated 2439-exNA Conc. ng/mL**	**Acc. Criterion ± 15 % of Nominal**	**Calculated 2439-exNA Conc. ng/mL**	**Acc. Criterion ± 15 % of Nominal**	**Calculated 2439-exNA Conc. ng/mL**	**Acc. Criterion ±15 % of Nominal**
AB03	12-Sep-2024	YE	HPLC-GLP-7	28, 35, 42, 49	1.0	100.0 %	2.9	96.7 %	453.6	100.8 %	896.4	99.6 %
				29, 36, 43, 50	0.9	90.0 %	3.0	100.0 %	456.5	101.4 %	900.8	100.1 %
				30, 37, 44, 51	0.9	90.0 %	3.0	100.0 %	451.5	100.3 %	897.2	99.7 %
				31, 38, 45, 52	1.0	100.0 %	2.9	96.7 %	451.1	100.2 %	890.5	98.9 %
				32, 39, 46, 53	0.9	90.0 %	2.9	96.7 %	452.7	100.6 %	897.9	99.8 %
				33, 40, 47, 54	1.0	100.0 %	2.8	93.3 %	448.6	99.7 %	895.8	99.5 %
Within-Run Accuracy/Precision III				**Mean**	0.95	95.0 %	2.92	97.2 %	452.33	100.5 %	896.43	99.6 %
				SD (±)	0.05		0.08		2.66		3.39	
				CV (%)	5.8 %		2.6 %		0.6 %		0.4 %	
				N	6		6		6		6	
Between-Run Accuracy/Precision				**Mean**	0.97	97.2 %	2.94	98.0 %	448.44	99.7 %	891.24	99.0 %
				SD (±)	0.06		0.07		6.78		5.95	
				CV (%)	5.9 %		2.4 %		1.5 %		0.7 %	
				N	18		18		18		18	

Table 15: Dilution Integrity for the Analysis of 2439-exNA in Dog Plasma

		Dilution 1:1,000	
		500,000.0 ng/mL	
Run ID	**Sample No.**	**Calculated 2439-exNA Conc. ng/mL**	**Acc. Criterion ± 15 % of Nominal**
AB01	66	478,617.2	95.7 %
	67	480,210.6	96.0 %
	68	475,289.1	95.1 %
	69	473,640.9	94.7 %
	70	477,554.6	95.5 %
	71	471,453.2	94.3 %
	Mean	476,127.60	95.2 %
	SD (±)	3,275.43	
	CV (%)	0.7 %	
	N	6	

Table 16: Long-Term Stability for the Analysis of 2439-exNA in Dog Plasma

Run ID	Validation Parameters	Sample No.	QC-Low		QC-High		QC-500xULOQ (Dilution: 1:1,000)	
			3.0 ng/mL		900.0 ng/mL		500,000.0 ng/mL	
			Calculated 2439-exNA Conc. ng/mL	Acc. Criterion ± 15 % of Nominal	Calculated 2439-exNA Conc. ng/mL	Acc. Criterion ± 15 % of Nominal	Calculated 2439-exNA Conc. ng/mL	Acc. Criterion ± 15 % of Nominal
AB01	T= 0 (-80 °C ± 20 °C)	13, 16, 19	2.6	86.7 %	812.0	90.2 %	418,440.0	83.7 %
		14, 17, 20	2.6	86.7 %	810.2	90.0 %	414,561.8	82.9 %
		15, 18, 21	2.6	86.7 %	809.9	90.0 %	420,672.2	84.1 %
		Mean	2.60	86.7 %	810.70	90.1 %	417,891.33	83.6 %
		SD (±)	0.00		1.14		3,091.93	
		CV (%)	0.0 %		0.1 %		0.7 %	
		N	3		3		3	
AB02	T= 0 (-80 °C ± 20 °C) Repetition of QC-500xULOQ	46	n.a.	n.a.	n.a.	n.a.	423,523.5	84.7 %
		47	n.a.	n.a.	n.a.	n.a.	398,005.4	79.6 %
		48	n.a.	n.a.	n.a.	n.a.	418,998.0	83.8 %
		Mean	n.a.	n.a.	n.a.	n.a.	413,508.97	82.7 %
		SD (±)	n.a.		n.a.		13,615.82	
		CV (%)	n.a.		n.a.		3.3 %	
		N	0		0		3	

Underlined QC sample / value did not fulfill the acceptance criterion but was included in the statistical calculations (Mean, SD and CV).

Table 16: Long-Term Stability for the Analysis of 2439-exNA in Dog Plasma (Continued)

Run ID	Validation Parameters	Sample No.	QC-Low		QC-High		QC-500xULOQ (Dilution: 1:1,000)	
			3.0 ng/mL		900.0 ng/mL		500,000.0900.0 ng/mL	
			Calculated 2439-exNA Conc. ng/mL	Acc. Criterion ± 15 % of Nominal	Calculated 2439-exNA Conc. ng/mL	Acc. Criterion ± 15 % of Nominal	Calculated 2439-exNA Conc. ng/mL	Acc. Criterion ± 15 % of Nominal
AB03	T= 0 (-80 °C ± 20 °C) newly prepared stability samples	18, 21, 24	2.7	90.0 %	853.5	94.8 %	451,505.7	90.3 %
		19, 22, 25	2.7	90.0 %	832.4	92.5 %	431,487.0	86.3 %
		20, 23, 26	2.6	86.7 %	846.5	94.1 %	432,270.4	86.5 %
		Mean	2.67	88.9 %	844.13	93.8 %	438,421.03	87.7 %
		SD (±)	0.06		10.75		11,338.42	
		CV (%)	2.2 %		1.3 %		2.6 %	
		N	3		3		3	

Table 17: Freeze-Thaw and Short-Term Stability for the Analysis of 2439-exNA in Dog Plasma

Run ID	Validation Parameters	Sample No.	QC-Low		QC-High		QC-500xULOQ (Dilution: 1:1,000)	
			3.0 ng/mL		900.0 ng/mL		500,000.0 ng/mL	
			Calculated 2439-exNA Conc. ng/mL	Acc. Criterion ± 15 % of Nominal	Calculated 2439-exNA Conc. ng/mL	Acc. Criterion ± 15 % of Nominal	Calculated 2439-exNA Conc. ng/mL	Acc. Criterion ± 15 % of Nominal
AB04	• FTS: 5 cycles (-80 °C ± 20 °C) • STS: 4 hours on ice	15, 18, 21	2.8	93.3 %	850.0	94.4 %	410,295.4	82.1 %
		16, 19, 22	2.9	96.7 %	851.7	94.6 %	439,025.0	87.8 %
		17, 20, 23	2.7	90.0 %	842.7	93.6 %	453,364.8	90.7 %
		Mean	2.80	93.3 %	848.13	94.2 %	434,228.40	86.8 %
		SD (±)	0.10		4.78		21,931.69	
		CV (%)	3.6 %		0.6 %		5.1 %	
		N	3		3		3	

Underlined QC sample / value did not fulfill the acceptance criterion but was included in the statistical calculations (Mean, SD and CV).

Table 18: Processed Sample Stability for the Analysis of 2439-exNA in Dog Plasma

Run ID	Date of Analysis	Sample No.	QC-Low 3.0 ng/mL		QC-High 900.0 ng/mL	
			Calculated 2439-exNA Conc. ng/mL	Acc. Criterion ± 15 % of Nominal	Calculated 2439-exNA Conc. ng/mL	Acc. Criterion ± 15 % of Nominal
AB02	10-Sep-2024 (Determination 1)	11, 14	2.9	96.7 %	889.9	98.9 %
		12, 15	2.8	93.3 %	895.3	99.5 %
		13, 16	2.7	90.0 %	899.5	99.9 %
		Mean	2.80	93.3 %	894.90	99.4 %
		SD (±)	0.10		4.81	
		CV (%)	3.6 %		0.5 %	
		N	3		3	
AB02	11-Sep-2024 (Determination 2)	50, 53	2.9	96.7 %	902.2	100.2 %
		51, 54	2.8	93.3 %	900.6	100.1 %
		52, 55	2.9	96.7 %	904.1	100.5 %
		Mean	2.87	95.6 %	902.30	100.3 %
		SD (±)	0.06		1.75	
		CV (%)	2.0 %		0.2 %	
		N	3		3	

Table 18: Processed Sample Stability for the Analysis of 2439-exNA in Dog Plasma (Continued)

Run ID	Date of Analysis	Sample No.	QC-Low 3.0 ng/mL		QC-High 900.0 ng/mL	
			Calculated 2439-exNA Conc. ng/mL	Acc. Criterion ± 15 % of Nominal	Calculated 2439-exNA Conc. ng/mL	Acc. Criterion ± 15 % of Nominal
AB03	12-Sep-2024 (Determination 3)	11*3, 14	10.2*3	340.0 %*3	913.6	101.5 %
		12, 15	2.9	96.7 %	921.1	102.3 %
		13, 16	2.9	96.7 %	929.2	103.2 %
		Mean	2.90	96.7 %	921.30	102.4 %
		SD (±)	0.00		7.80	
		CV (%)	0.0 %		0.8 %	
		N	2		3	
AB03	13-Sep-2024 (Determination 4)	58, 61	3.0	100.0 %	925.0	102.8 %
		59, 62	2.9	96.7 %	922.4	102.5 %
		60, 63	2.9	96.7 %	936.8	104.1 %
		Mean	2.93	97.8 %	928.07	103.1 %
		SD (±)	0.06		7.67	
		CV (%)	2.0 %		0.8 %	
		N	3		3	

*3 Sample / value was excluded from the calculation of mean, SD and CV: single value could be designated as an outlier at a probability of at least 95% according to W. Gottwald, Statistik für Anwender, Wiley-VCH Verlag.

Table 19: Short-Term Stability of Stock and Working Solutions for the Analysis of 2439-exNA in Dog Plasma

Run ID	Validation Parameters	Sample No.	Working Solution LLOQ		Working Solution ULOQ		Stock Solution (diluted to QC-High)	
			1.0 ng/mL		1,000.0 ng/mL		900.0 ng/mL	
			Calculated 2439-exNA Conc. ng/mL	Acc. Criterion ± 20 % of Nominal	Calculated 2439-exNA Conc. ng/mL	Acc. Criterion ± 15 % of Nominal	Calculated 2439-exNA Conc. ng/mL	Acc. Criterion ± 15 % of Nominal
AB01	4 hours at RT	73, 76, 79	0.9	90.0 %	952.2	95.2 %	895.0	99.4 %
		74, 77, 80	1.0	100.0 %	957.0	95.7 %	889.6	98.8 %
		75, 78, 81	1.0	100.0 %	956.2	95.6 %	892.1	99.1 %
		Mean	0.97	96.7 %	955.13	95.5 %	892.23	99.1 %
		SD (±)	0.06		2.57		2.70	
		CV (%)	6.0 %		0.3 %		0.3 %	
		N	3		3		3	

Table 20: Calibration Curve Parameters for the Analysis of 2439-exNA in Dog Plasma (Other Validation Runs)

Equation for the calibration curve using a linear regression model weighted 1/X: y = m*x + b					
Run ID	**Date of Analysis**	**Sample No.**	**Slope [m]**	**Y-Offset [b]**	**R-Squared**
AN402.0-24_01_AB04	18-Sep-2024	3 – 8, 30 – 35	0.03687	-0.00427	0.99996

Table 21: Results of the Calibration Standards for the Analysis of 2439-exNA in Dog Plasma (Other Validation Runs)

		Cal-Level-2 (LLOQ)		**Cal-Level-3**		**Cal-Level-4**		**Cal-Level-5**		**Cal-Level-6**		**Cal-Level-7 (ULOQ)**	
		1.0 ng/mL		**5.0 ng/mL**		**50.0 ng/mL**		**100.0 ng/mL**		**500.0 ng/mL**		**1,000.0 ng/mL**	
Run ID	**Sample No.**	**Calculated 2439-exNA Conc. ng/mL**	**Acc. Criterion ± 20 % of Nominal**	**Calculated 2439-exNA Conc. ng/mL**	**Acc. Criterion ± 15 % of Nominal**	**Calculated 2439-exNA Conc. ng/mL**	**Acc. Criterion ± 15 % of Nominal**	**Calculated 2439-exNA Conc. ng/mL**	**Acc. Criterion ± 15 % of Nominal**	**Calculated 2439-exNA Conc. ng/mL**	**Acc. Criterion ± 15 % of Nominal**	**Calculated 2439-exNA Conc. ng/mL**	**Acc. Criterion ± 15 % of Nominal**
AB04	3 – 8	1.0	100.0 %	4.9	98.0 %	49.9	99.8 %	99.7	99.7 %	499.5	99.9 %	989.1	98.9 %
AB04	30 – 35	1.0	100.0 %	5.0	100.0 %	50.4	100.8 %	102.3	102.3 %	507.3	101.5 %	1,001.9	100.2 %
	Mean	1.00	100.0 %	4.95	99.0 %	50.15	100.3 %	101.00	101.0 %	503.40	100.7 %	995.50	99.6 %
	SD (±)	0.00		0.07		0.35		1.84		5.52		9.05	
	CV (%)	0.0 %		1.4 %		0.7 %		1.8 %		1.1 %		0.9 %	
	N	2		2		2		2		2		2	

Table 22: Results of the QC Samples for the Analysis of 2439-exNA in Dog Plasma (Other Validation Runs)

		QC-Low		QC-Mid		QC-High	
		3.0 ng/mL		450.0 ng/mL		900.0 ng/mL	
Run ID	**Sample No.**	**Calculated 2439-exNA Conc. ng/mL**	**Acc. Criterion ± 15 % of Nominal**	**Calculated 2439-exNA Conc. ng/mL**	**Acc. Criterion ± 15 % of Nominal**	**Calculated 2439-exNA Conc. ng/mL**	**Acc. Criterion ± 15 % of Nominal**
AB04	11 – 13	3.1	103.3 %	459.0	102.0 %	905.1	100.6 %
	25 – 27	3.2	106.7 %	459.1	102.0 %	908.4	100.9 %
	Mean	3.15	105.0 %	459.05	102.0 %	906.75	100.8 %
	SD (±)	0.07		0.07		2.33	
	CV (%)	2.2 %		0.0 %		0.3 %	
	N	2		2		2	

Table 23: Selectivity and Carry-Over for the Analysis of 2439-exNA in Dog Plasma (Other Validation Runs)

		Blank Sample for Selectivity (Cal-Level 1)			Cal-Level-2 (LLOQ) 1.0 ng/mL			Blank Sample for Carry-Over		
Run ID	**Sample No.**	**Peak Area [mV*min]**	**Calculated 2439-exNA Conc. ng/mL**	**Acc. Criterion ≤ 20 % of LLOQ**	**Peak Area [mV*min]**	**Calculated 2439-exNA Conc. ng/mL**	**Acc. Criterion +/- 20 % of Nominal**	**Peak Area [mV*min]**	**Calculated 2439-exNA Conc. ng/mL**	**Acc. Criterion ≤ 20 % of LLOQ**
AB04	2, 3, 9	0.0000	BQL	0.0 %	0.0308	1.0	100.0 %	0.0000	BQL	0.0 %
AB04	29, 30, 36	0.0000	BQL	0.0 %	0.0337	1.0	100.0 %	0.0000	BQL	0.0 %

14 SUPPLEMENTARY INFORMATION

14.1 Supplemental File 1

Axolabs *MDR_2439exNA_DogPlasma*

14.2 Supplemental File 2

Axolabs *STM_Broad_2439-exNA_DogPlasma*